

BLAST ANALYSIS REPORT: Human Insulin

Protein Analyzed: Human insulin (AAA59172.1) is a 110 amino acid peptide hormone that regulates glucose metabolism. It's one of the most conserved proteins across mammals, making it ideal for homology analysis.

BLAST Results Summary: The BLAST search against the NCBI nr database identified highly significant matches across primate and non-primate species. The top 5 homologous sequences showed exceptionally high sequence identity (97–99%), indicating strong evolutionary conservation.

Top 5 Hits:

1. Pan troglodytes (Chimpanzee) insulin preproprotein — 98.18% identity, E-value: $1e-72$
2. Pan paniscus (Bonobo) insulin — 99.09% identity, E-value: $2e-72$
3. Macaca nemestrina (Pig-tailed macaque) insulin isoform X1 — 98.18% identity, E-value: $4e-72$
4. Chlorocebus sabaeus (Green monkey) insulin isoform X2 — 97.27% identity, E-value: $5e-71$
5. Chlorocebus sabaeus insulin isoform X1 — 97.27% identity, E-value: $7e-70$

Interpretation: The extremely low E-values (< 0.01) indicate statistically significant alignments. The high percent identity across primates and other mammals demonstrates insulin's evolutionary importance in glucose homeostasis. Cross-species conservation suggests functional constraints on this protein.

Screenshot:



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Protein [Advanced](#)

FASTA ▾

insulin [Homo sapiens]

GenBank: AAA59172.1
[GenPept](#) [Identical Proteins](#) [Graphics](#)

>AAA59172.1 insulin [Homo sapiens]
MALWMRLRLPLALLALWGPDPAAFVNQHLGSHLVEALYLVCGERGFFYTPKTRREAEDLQVGQVELGG
GPGAGSLQPLALEGSLQKRGIVEQCCTSIICSLYQLENYCN

Figure 1: FASTA sequence of Human Insulin



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AAA59172.1 insulin [Homo sapiens]

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Database
nr_cluster_seq [See details ▼](#)

Query ID
lcl|Query_5424797

Description
AAA59172.1 insulin [Homo sapiens]

Molecule type
amino acid

Query Length
110

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Organism only top 20 will appear ☐ exclude

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Percent Identity

to

E value

to

Query Coverage

to

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Descriptions
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Alignments
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Sequences producing significant alignments


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Sequences producing significant alignments												Manage Column		View							
☐ select all 0 sequences selected												GenPept		Graphics		Distance tree of results		Multiple alignment		MSA Viewer	
	Description	Scientific Name	Common Name	Taxid	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession										
☐	insulin preproprotein [Pan troglodytes]	Pan tro...	chimpan...	9598	221	221	100%	1e-72	98.18%	110	NP_001008996.1										
☐	insulin [Pan paniscus]	Pan pan...	pygmy c...	9597	224	224	100%	2e-72	99.09%	204	XP_034787832.1										
☐	insulin isoform X1 [Macaca nemestrina]	Macaca ...	pig-tail...	9545	221	221	100%	4e-72	98.18%	153	XP_011719619.2										
☐	insulin isoform X2 [Chlorocebus sabaeus]	Chloroc...	green m...	60711	218	218	100%	5e-71	97.27%	144	XP_008002825.1										
☐	insulin isoform X1 [Chlorocebus sabaeus]	Chloroc...	green m...	60711	218	218	100%	7e-70	97.27%	209	XP_008002752.1										
☐	INS isoform 5 [Pongo abelii]	Pongo a...	Sumatra...	9501	191	191	100%	1e-60	88.18%	98	PNJ16423.1										
☐	insulin [Tupaia tana]	Tupaia t...	large tre...	70687	184	184	100%	4e-58	90.00%	110	KAN7991649.1										
☐	insulin [Micromys minutus]	Microm...	Europea...	13151	184	184	100%	5e-58	79.09%	110	XP_080768357.1										
☐	ZZ-proinsulin [Cloning vector pET28a_ZZ_proinsulin]	Cloning...NA	2899524	183	183	78%	2e-56	100.00%	210	UFG90408.1											
☐	insulin [Rhinolophus affinis]	Rhinolo...	interme...	59477	178	178	100%	2e-55	88.18%	110	KAN3685427.1										
☐	hypothetical protein [Streptomyces sp. Lzd4kr]	Strepto...	NA	3014556	179	179	78%	2e-55	100.00%	143	MCZ4612290.1										

Figure 3: top 5 matches with their names, identity %, and E-values